

Subrahmanya (Raju) Krishnamoorthy, Ph.D.

✉ krishnamoorthy@alum.mit.edu  [notnotraju](https://github.com/notnotraju)  [rkrishnamoorthy](https://orcid.org/rkrishnamoorthy)
 [rajukrishnamoorthy](https://www.linkedin.com/in/rajukrishnamoorthy)  [notnotraju.github.io](https://github.com/notnotraju)

Employment History

- 2025.06 – present  **Cryptographic Engineer**, Aztec Labs.
- 2024.07 – 2024.12  **Cryptographer**, Irreducible.
- 2022.10 – 2024.06  **Wissenschaftlicher Mitarbeiter**, Humboldt Universität Berlin.
- 2020.10 – 2022.10  **Wissenschaftlicher Mitarbeiter**, Bergische Universität Wuppertal
- 2018.08 – 2020.08  **Limited Term Assistant Professor** University of Georgia, Athens
- 2016.08 – 2018.08  **NSF Postdoctoral Fellow**, Freie Universität Berlin, **Supervisor**: Hélène Esnault

Education

- 2010 – 2016  **Ph.D., Columbia University** Mathematics.
Thesis title: *Dynamics, Graph Theory, and Barsotti-Tate Groups: Variations on a Theme of Mochizuki*. Supervisor: Johan de Jong
- 2005 – 2008  **B.S., MIT** Mathematics with Computer Science

Research Publications and Preprints

Journal Articles

- 1 R. Krishnamoorthy, J. Yang, and K. Zuo, “A Lefschetz theorem for crystalline representations,” *Algebra Number Theory*, 2025, to appear. arXiv: [2003.08906](https://arxiv.org/abs/2003.08906) [math.AG].
- 2 R. Krishnamoorthy and M. Sheng, “Periodicity of Hitchin’s uniformizing Higgs bundles,” *Int. Math. Res. Not.*, vol. 2024, no. 11, pp. 9440–9468, Mar. 2024, ISSN: 1073-7928.  DOI: [10.1093/imrn/rnae042](https://doi.org/10.1093/imrn/rnae042).
- 3 R. Krishnamoorthy, J. Yang, and K. Zuo, “Constructing abelian varieties from rank 2 Galois representations,” *Compos. Math.*, vol. 160, no. 4, pp. 709–731, 2024.  DOI: [10.1112/S0010437X23007728](https://doi.org/10.1112/S0010437X23007728).
- 4 R. Krishnamoorthy, “Rank 2 local systems, Barsotti-Tate groups, and Shimura curves,” *Algebra Number Theory*, vol. 16, no. 2, pp. 231–259, 2022, ISSN: 1937-0652.  DOI: [10.2140/ant.2022.16.231](https://doi.org/10.2140/ant.2022.16.231).
- 5 R. Krishnamoorthy and A. Pál, “Rank 2 local systems and abelian varieties. II,” *Compos. Math.*, vol. 158, no. 4, pp. 868–892, 2022, ISSN: 0010-437X.  DOI: [10.1112/S0010437X22007333](https://doi.org/10.1112/S0010437X22007333).
- 6 R. Krishnamoorthy and A. Pál, “Rank 2 local systems and abelian varieties,” *Sel. Math., New Ser.*, vol. 27, no. 4, p. 40, 2021, Id/No 51, ISSN: 1022-1824.  DOI: [10.1007/s00029-021-00669-8](https://doi.org/10.1007/s00029-021-00669-8).
- 7 R. Krishnamoorthy, “Correspondences without a core,” *Algebra Number Theory*, vol. 12, no. 5, pp. 1173–1214, 2018, ISSN: 1937-0652.  DOI: [10.2140/ant.2018.12.1173](https://doi.org/10.2140/ant.2018.12.1173).
- 8 R. C. Daileda, R. Krishnamoorthy, and A. Malyshev, “Maximal class numbers of CM number fields,” *J. Number Theory*, vol. 130, no. 4, pp. 936–943, 2010, ISSN: 0022-314X.  DOI: [10.1016/j.jnt.2009.09.013](https://doi.org/10.1016/j.jnt.2009.09.013).

Preprints

- 1 R. Krishnamoorthy and Y. H. J. Lam, *Constructing abelian varieties from rank 3 galois representations with real trace field*, 2024. arXiv: [2403.18138](https://arxiv.org/abs/2403.18138) [math.AG].
- 2 R. Krishnamoorthy and Y. H. J. Lam, *Frobenius trace fields of cohomologically rigid local systems*, 2023. arXiv: [2308.10642](https://arxiv.org/abs/2308.10642) [math.AG].

- 3 P. Engel, R. Krishnamoorthy, and D. Litt, *The Manin-Mumford conjecture in genus 2 and rational curves on $K3$ surfaces*, 2022. arXiv: [2208.08729 \[math.AG\]](#).
- 4 R. Krishnamoorthy and M. Sheng, *Periodic de Rham bundles over curves*, 2022. arXiv: [2011.03268 \[math.AG\]](#).
- 5 R. Krishnamoorthy, J. Yang, and K. Zuo, *Deformation theory of periodic Higgs-de Rham flows*, 2020. arXiv: [2005.00579 \[math.AG\]](#).
- 6 R. Krishnamoorthy, J. Yang, and K. Zuo, *Finiteness of logarithmic crystalline representations*, 2020. arXiv: [2005.13472 \[math.AG\]](#).
- 7 R. Krishnamoorthy, J. Yang, and K. Zuo, *Finiteness of logarithmic crystalline representations II*, 2020. arXiv: [2009.00074 \[math.AG\]](#).

Skills

Programming	Python, Rust, C++, SageMath, Circom, Halo2.
Cryptography	Experienced with zero-knowledge proofs, succinct verifiable computation, and their synthesis: zk-SNARKS. Knowledgeable about the complexity-theoretic foundations.
Mathematics	Expert in algebraic geometry over arithmetic fields, in particular elliptic curves and higher-dimensional abelian varieties over finite fields.

Selected Computer Science Experience

Personal	I have implemented a variety of cryptographic and verifiable computation algorithms in Python and in Rust, including GKR , batch IPA prover in Rust and verifier in halo2 (detailed explanation here).
Irreducible	Among my software contributions , I implemented ring-switching (small-to-large field reduction for multilinear PCS), black-box batching of multilinear PCS, and non-two-primary binary-field FFT extrapolation . I also open-sourced binus-models , a set of Python models of core protocols in Binius. Theoretically, I contributed several novel insights to FRI-Binius .

Other academic experience

Teaching

2008 – 2024	Have taught a variety of undergraduate classes (in English and German) and have run many graduate/research level seminars. More details may be found here .
2009 – 2010	Co-started a creative math class for kids through sprout (in Somerville, MA).

Talks

2009 – 2024	Have given numerous invited seminar and conference talks on my research in Canada, China, France, Germany, the Netherlands, Poland, and the United States.
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